

**REPORT ON: SOFTWARE REQUIREMENTS  
SPECIFICATION**

**COURSE CODE: CSE-404**  
**COURSE TITLE: SOFTWARE ENGINEERING AND ISD  
LABORATORY**

Submitted by

Md. Al-Mizan Akon (349)

Nirob Mondal (357)

Najmul Islam Jesan (363)

Aung Sing Marma (377)

Jyoti Pal (391)

Fateema Binti Taher Eva(403)

Submitted to

Dr. Md. Musfique Anwar, Professor

Dr. Md. Humayun Kabir, Professor



Computer Science and Engineering  
Jahangirnagar University

Dhaka, Bangladesh

July 05, 2026

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# 1. Introduction

## 1.1 Purpose

The purpose of this document is to provide a comprehensive **Software Requirements Specification (SRS)** for the **Smart Department**—a web-based platform designed to digitize and streamline the academic operations of the Department of Computer Science and Engineering (CSE) at **Jahangirnagar University, Savar, Dhaka-1342, Bangladesh**.

Currently, the department manages its academic scheduling, classroom allocation, class tracking, Class Test (CT) scheduling, result management, and resource distribution through manual or semi-manual processes. These traditional methods often lead to scheduling conflicts, inefficient resource utilization, communication gaps between faculty members and students, and the absence of a centralized academic information management system.

The proposed system aims to address these challenges by providing a unified digital platform with role-based access control. Through this platform:

- Administrators can allocate classrooms and laboratories for the entire semester while avoiding scheduling conflicts.
- Teachers can manage class schedules, attendance, class tests, assignments, and assessment records.
- Class Representatives (CRs) can upload academic resources, notices, and class test results for students.
- Students can access their personalized class schedules, classroom information, examination results, CT marks, and downloadable study materials from a single portal.

The system is designed to improve academic coordination, enhance communication among stakeholders, optimize resource utilization, and ensure efficient management of

departmental academic activities.

This document defines the functional and non-functional requirements of the system, identifies the stakeholders and their respective needs, outlines the operating environment, and specifies the assumptions and constraints under which the system will be developed, deployed, and maintained.

## 1.2 Intended Audience

### 1.2.1 Intended Audience and Reading Suggestions

This SRS addresses multiple stakeholder groups involved in the design, development, deployment, and operation of **Smart Department**. Each group is advised to focus on the sections most relevant to their role, as summarised in Table 1.1.

| <b>Audience</b>                 | <b>Role / Primary Interest</b>                          | <b>Suggested Sections</b>                                                                                               |
|---------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Project Manager / Sponsor       | Evaluate feasibility, scope, and resource allocation    | Section 1.1, Section 1.4, Section 2.2, Section 2.3                                                                      |
| System Analyst / Designer       | Translate requirements into system architecture         | All sections; priority focus on Section 3, Section 4, Section 5                                                         |
| Developer                       | Implement functional and non-functional requirements    | Section 3 (Functional Requirements), Section 4 (External Interfaces), Section 5 (Non-Functional Requirements)           |
| Tester / QA Engineer            | Design test cases and acceptance criteria               | Section 3 (all functional requirements), Section 5 (performance, security, quality attributes)                          |
| Department Admin (Operations)   | Understand system capabilities; oversee deployment      | Section 2.3 User Classes, Section 3.1 Authentication, Section 3.13 Conflict Detection, Section 3.14 Semester Promotion  |
| End Users – Students & Teachers | Understand features relevant to daily academic workflow | Section 2.3 User Classes, Section 3.1 (login portion); see also the separate <i>User Manual</i>                         |
| Future Maintainers              | Extend or modify the system after initial deployment    | All sections; priority focus on Section 3, Section 4, Section 5, Appendix C (Data Dictionary), Appendix D (Open Issues) |

Table 1.1: Intended audience and reading guide.

### 1.3 Intended Use

| Aspect             | Description                                                                        |
|--------------------|------------------------------------------------------------------------------------|
| Routine Management | Create, update, and publish class routines efficiently.                            |
| Scheduling         | Manage examination schedules and classroom allocations while minimizing conflicts. |
| Academic Resources | Share assignments, course materials, and academic notices.                         |
| Communication      | Provide real-time notifications and announcements to users.                        |
| Access Control     | Offer secure, role-based access for administrators, teachers, CRs, and students.   |
| Overall Objective  | Improve efficiency, accuracy, and transparency in academic routine management.     |

Table 1.2: Intended Use Smart Department

### 1.4 Product Scope

The **Smart Department** is a web-based platform designed to digitize and simplify academic scheduling within the Department of Computer Science and Engineering, Jalpaiguri University.

**The system aims to:**

- Generate and manage class routines efficiently.
- Schedule examinations and allocate classrooms.
- Publish assignments, notices, and academic resources.
- Reduce scheduling conflicts through centralized management.
- Provide real-time access to academic information.
- Improve communication among students, teachers, CRs, and administrators.

### 1.5 Risk Definitions

| Risk ID | Risk Description                                                                                                                                                                                                                      | Severity | Likelihood | Mitigation Strategy                                                                                                                                                                                                                                                                          |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R-01    | <b>Scheduling Conflict:</b><br>A teacher reschedules a class to a time slot where the assigned classroom or lab is already occupied by another class, exam, or event.                                                                 | High     | High       | The system will implement a real-time conflict detection engine that checks room availability, teacher availability, and batch schedule before allowing any schedule change. Conflicting requests will be blocked with a descriptive error message.                                          |
| R-02    | <b>Unauthorized Schedule Modification:</b> A teacher modifies another teacher's class schedule using a shared or compromised account.                                                                                                 | High     | Medium     | Each teacher will be assigned a unique ID. The system will enforce ownership-based access control — only the teacher assigned to a class can modify that specific class's schedule. All modifications will be logged with timestamps and user IDs for audit.                                 |
| R-03    | <b>Data Loss During Semester Transition:</b><br>When an admin promotes a batch to the next semester, the previous semester's routine data is deleted. If the promotion is premature or erroneous, critical schedule data may be lost. | High     | Low        | The system will require admin confirmation before semester promotion. A soft-delete mechanism will archive the previous semester's data for a configurable retention period (e.g., 30 days) before permanent deletion. CR promotion requests will undergo validity checks before processing. |



| Risk ID | Risk Description                                                                                                                                                                | Severity | Likelihood | Mitigation Strategy                                                                                                                                                                                                                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R-04    | <b>Incorrect User Registration:</b> A student or teacher registers with false credentials, gaining unauthorized access to the system and its data.                              | Medium   | Medium     | Registration will require admin-verified data. The admin will pre-load authorized student and teacher information (university ID, email, batch, etc.) into the system. During registration, user-provided details will be cross-verified against this pre-loaded dataset. Only verified users will be granted access. |
| R-05    | <b>System Downtime During Peak Usage:</b> The system experiences high traffic during exam routine publication, result uploads, or semester start, causing slowdowns or outages. | Medium   | Medium     | The application will be deployed on a scalable cloud infrastructure. Database queries will be optimized with proper indexing. Caching mechanisms will be used for frequently accessed data (e.g., routines, results). Load testing will be conducted before deployment.                                               |
| R-06    | <b>Role Escalation by CR:</b> A Class Representative misuses their elevated privileges to upload incorrect results or tamper with shared resources.                             | Medium   | Low        | All CR actions (result uploads, resource uploads, promotion requests) will be logged and auditable. The admin will have the ability to review, revert, or override any CR action. Sensitive operations will require additional confirmation steps.                                                                    |

| Risk ID | Risk Description                                                                                                                                          | Severity | Likelihood | Mitigation Strategy                                                                                                                                                                                                                  |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R-07    | <b>Browser Compatibility Issues:</b> The web application may not render correctly across all browsers and devices, leading to a degraded user experience. | Low      | Medium     | The frontend will be built with React, which provides cross-browser compatibility. The application will be tested on Chrome, Firefox, Safari, Edge, and Brave. Responsive design will ensure usability on mobile and tablet devices. |

## 2. Overall Description

### 2.1 User Classes and Characteristics

The system defines four distinct user classes, each with different access levels, responsibilities, and interaction patterns. All users belong to the CSE department of Jahangirnagar University.

#### 2.1.1 Normal Student

##### **Description:**

Normal Students are the primary consumers of the system. They are undergraduate (Honours) or postgraduate (Masters) students currently enrolled in the CSE department. Each student is associated with a specific batch and a current semester. The system provides them with a personalized view of their academic schedule, upcoming assessments, marks, and downloadable resources.

Due to the nature of the department's academic progression, not all students follow a linear path through their semesters. The system must accommodate the following student sub-categories, all managed by the Admin:

##### **Student Sub-Categories:**

| Sub-Category                 | Description                                                                                                                                                                                                                                                                                                                              |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Regular Student</b>       | A student who has never dropped out, taken a year off, or failed a course. They progress linearly through semesters with their original batch.                                                                                                                                                                                           |
| <b>Year-Drop Student</b>     | A student who has taken one or more years off and subsequently resumed their studies with a junior batch (e.g., 1–2 batches behind their original). The Admin updates their batch and semester assignment accordingly.                                                                                                                   |
| <b>Discontinued Student</b>  | A student who has permanently left the program. The Admin marks their status as "Discontinued," and they lose access to personalized dashboard features. Their historical data (marks, enrollment) is retained for records.                                                                                                              |
| <b>Course-Retake Student</b> | A student who has failed a particular course and must retake the semester final exam with another batch that is currently offering that course. During the retake period, the Admin grants the student temporary multi-batch access so they can view the exam routine and relevant schedule of the other batch in addition to their own. |

**Note:** In special cases (e.g., course retake), a single student may require access to multiple batches simultaneously. The Admin manages all such assignments — updating a student's semester, batch, status, or granting temporary multi-batch visibility as needed.

#### Characteristics:

| Attribute                     | Detail                                                                                                                                                                              |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Program Types</b>          | Honours (B.Sc.) — 4-year program with 8 semesters; Masters (M.Sc.) — program with 2 semesters                                                                                       |
| <b>Active Honours Batches</b> | 5–6 batches may be active at any time due to irregular semester completion timelines across batches (e.g., 50th, 51st, 52nd, 53rd, 54th, 55th)                                      |
| <b>Active Masters Batches</b> | 1 or more batches may be active at any time                                                                                                                                         |
| <b>Enrollment</b>             | Students are enrolled in their current semester by default. They see their own batch's information. In special cases (course retake), Admin may grant access to additional batches. |

| Attribute                    | Detail                                                                            |
|------------------------------|-----------------------------------------------------------------------------------|
| <b>Technical Proficiency</b> | Moderate to high (CSE students are familiar with web applications)                |
| <b>Frequency of Use</b>      | Daily — to check class schedules, upcoming CTs, assignments, and marks            |
| <b>Access Level</b>          | Read-only for personalized data; read access to public pages (resources, results) |

### 2.1.2 Class Representative (CR)

#### Description:

Class Representatives are students who have been designated (by the Admin) to serve as the liaison between their batch and the department. Each batch has **two CRs — one male and one female** — who share equal responsibilities and privileges. They possess all the capabilities of a Normal Student plus additional elevated privileges for managing resources and results on behalf of their batch. CRs may change each semester as decided by the department, and the Admin manages all CR appointments and removals.

#### Characteristics:

| Attribute                    | Detail                                                                                                                                              |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Selection</b>             | Appointed by the Admin. A student can be promoted to CR, and a CR can be demoted back to a Normal Student by the Admin.                             |
| <b>Count Per Batch</b>       | Each active batch has exactly <b>2 CRs</b> (1 male, 1 female). Both CRs have identical privileges and can perform the same actions for their batch. |
| <b>Semester-Wise Change</b>  | CRs may change at the start of each semester. The Admin is responsible for updating CR assignments when transitions occur.                          |
| <b>Scope</b>                 | The CRs' elevated actions apply only to their own batch.                                                                                            |
| <b>Technical Proficiency</b> | High (expected to be more engaged with the system than regular students)                                                                            |
| <b>Frequency of Use</b>      | Several times per week — for uploading resources, results, and managing semester transitions                                                        |

| Attribute           | Detail                                                                                                                                                                                                                                                                                   |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Access Level</b> | All Normal Student permissions + upload resources to the public Resource Page, upload semester final results to the public Result Page, submit semester promotion requests for their batch, and submit class cancellation/reschedule requests to course teachers on behalf of the batch. |

### 2.1.3 Teacher

#### Description:

Teachers are faculty members of the CSE department who use the system to manage their classes, assessments, and academic events. A teacher may be involved with multiple batches across different semesters simultaneously. Each teacher is assigned a unique system ID to ensure accountability and prevent unauthorized modification of other teachers' schedules.

The department has a **Chairman** role, which is held by one teacher at a time. The Chairman is a teacher with elevated administrative privileges (e.g., allocating seminars and workshops). The Chairman role is **rotational** — when a teacher's tenure as Chairman ends, they transition back to a regular Teacher role, and another teacher is assigned as Chairman. The Admin manages this role transition within the system.

#### Characteristics:

| Attribute                      | Detail                                                                                                                                                                                                                                                          |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Multi-Batch Involvement</b> | A teacher may teach courses in multiple batches simultaneously                                                                                                                                                                                                  |
| <b>Unique ID</b>               | Each teacher has a unique system-generated ID for tracking and access control                                                                                                                                                                                   |
| <b>Chairman Role</b>           | One teacher holds the Chairman role at any time. The Chairman has additional privileges to allocate seminars and workshops. When the tenure ends, the Admin reassigns the role to another teacher, and the previous Chairman reverts to a regular Teacher role. |
| <b>Technical Proficiency</b>   | Moderate to high                                                                                                                                                                                                                                                |
| <b>Frequency of Use</b>        | Daily — to check schedules, manage classes, upload CT marks, assign assignments                                                                                                                                                                                 |

| Attribute           | Detail                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Access Level</b> | Personalized dashboard (multi-batch view), class cancellation/rescheduling (own classes only with conflict check), CT scheduling (within own class times), assignment creation, CT marks upload, and review/approval of CR-submitted class cancellation/reschedule requests. The Chairman additionally has privileges to allocate rooms and time slots for seminars and workshops. |

### 2.1.4 Admin

#### Description:

Admins are the system superusers who manage the foundational data and configurations of the platform. They are responsible for setting up semesters, creating day-wise class routines, managing physical resources (classrooms, labs), handling user verification and registration data, managing holidays, generating exam routines, processing semester promotions, managing student sub-category transitions, and assigning the Chairman role among teachers. The Admin role is typically assigned to designated department staff or faculty authorized by the department Chairman.

#### Characteristics:

| Attribute                    | Detail                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibilities</b>      | Full system configuration and management                                                                                                                                                                                                                                                                                                                                                            |
| <b>Technical Proficiency</b> | High (trained on the system)                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Frequency of Use</b>      | Daily during semester setup; weekly during regular operations; high activity during semester transitions and exam periods                                                                                                                                                                                                                                                                           |
| <b>Access Level</b>          | Full access to all system modules. Can view all data across all batches and semesters. Manages user registration data, semester creation, routine generation, resource allocation, holiday management, exam routine generation, semester promotions, role assignments (Student ↔ CR, Teacher ↔ Chairman), student sub-category management (batch reassignment, multi-batch access, status updates). |

## 2.2 User Needs

This section describes the specific needs of each user class that the system must fulfill. Each need is assigned a unique identifier and a priority level based on its importance to the system.

### 2.2.1 Normal Student Needs

Table 2.6: Normal Student Needs

| Need ID | Need Description                                                                                                                                                                                                        | Priority |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| SN-01   | <b>Personalized Class Schedule:</b> View a calendar-based, day-wise schedule for the current semester showing class name, room number, time slot, and course teacher, filtered to display only the student's own batch. | High     |
| SN-02   | <b>Upcoming CT Notifications:</b> View a list of upcoming Class Tests (CTs) with date, time, course name, and room number.                                                                                              | High     |
| SN-03   | <b>Upcoming Assignment Tracker:</b> View a list of pending assignments including course name, description, and due date.                                                                                                | High     |
| SN-04   | <b>CT Marks Viewer:</b> View marks obtained in all Class Tests for the current semester, organized by course.                                                                                                           | High     |
| SN-05   | <b>Class Count Tracker:</b> View the total number of classes conducted per course by each teacher during the current semester.                                                                                          | Medium   |
| SN-06   | <b>Resource Download (Public Page):</b> Access a public repository of study materials (notes, slides, past papers) categorized by academic year and semester, and download available resources.                         | Medium   |
| SN-07   | <b>Semester Final Result (Public Page):</b> Access published semester final results for all students, with the most recently published results displayed first.                                                         | Medium   |

### 2.2.2 Class Representative (CR) Needs

Table 2.7: Class Representative Needs

| Need ID | Need Description                                                                                                                                        | Priority |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| CN-01   | <b>All Normal Student Needs:</b> A Class Representative possesses all functionalities available to a Normal Student (SN-01 to SN-07).                   | High     |
| CN-02   | <b>Resource Upload:</b> Upload study resources (PDFs, documents, images) to the public Resource Page categorized by academic year and semester.         | High     |
| CN-03   | <b>Result Upload:</b> Upload semester final examination results for the assigned batch to the public Result Page.                                       | High     |
| CN-04   | <b>Semester Promotion Request:</b> Submit a request to the Admin to promote the batch to the next semester after completion of all academic activities. | High     |



### 2.2.3 Teacher Needs

Table 2.8: Teacher Needs

| Need ID | Need Description                                                                                                                                                            | Priority |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| TN-01   | <b>Multi-Batch Dashboard:</b> View a consolidated dashboard displaying all upcoming classes across every assigned batch.                                                    | High     |
| TN-02   | <b>Batch-Wise Schedule View:</b> Access detailed information for a selected batch, including upcoming classes, class history, and total class count.                        | High     |
| TN-03   | <b>Class Cancellation:</b> Cancel a scheduled class. The system updates the schedule and notifies affected students.                                                        | High     |
| TN-04   | <b>Class Time Update:</b> Modify the time of a scheduled class within the same day while ensuring no scheduling conflicts.                                                  | High     |
| TN-05   | <b>Class Reassignment:</b> Reschedule a class to another day. The system checks room, teacher, and batch availability and suggests valid time slots.                        | High     |
| TN-06   | <b>CT Scheduling:</b> Schedule a Class Test during an existing class slot, automatically converting that class session into a CT.                                           | High     |
| TN-07   | <b>CT Marks Upload:</b> Upload Class Test marks for students enrolled in the selected batch.                                                                                | High     |
| TN-08   | <b>Assignment Creation:</b> Create assignments for a batch by specifying a description and submission deadline.                                                             | Medium   |
| TN-09   | <b>Seminar/Workshop Allocation:</b> Allocate rooms and time slots for seminars or workshops (restricted to the Department Chairman). The system performs conflict checking. | Medium   |
| TN-10   | <b>Class Count View:</b> View the total number of classes conducted for each assigned batch during the current semester.                                                    | Low      |

### 2.2.4 Admin Needs

Table 2.9: Administrator Needs

| Need ID | Need Description                                                                                                                                                               | Priority |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| AN-01   | <b>Student Data Pre-loading:</b> Import or manually enter student information for authentication and registration verification.                                                | High     |
| AN-02   | <b>Teacher Data Pre-loading:</b> Import or manually enter teacher information for authentication and registration verification.                                                | High     |
| AN-03   | <b>Semester Creation:</b> Create a semester by specifying start date, end date, associated batch, and course list.                                                             | High     |
| AN-04   | <b>Day-Wise Routine Generation:</b> Generate the complete semester schedule from the department routine by assigning teachers, classrooms, and time slots.                     | High     |
| AN-05   | <b>Classroom and Lab Resource Allocation:</b> Allocate classrooms, laboratories, and multipurpose rooms while preventing scheduling conflicts.                                 | High     |
| AN-06   | <b>Semester Final Exam Routine:</b> Generate and publish the semester final examination schedule with room and time allocations.                                               | High     |
| AN-07   | <b>Semester Promotion Processing:</b> Review CR requests, validate completion requirements, promote the batch to the next semester, and clear the previous semester's routine. | High     |
| AN-08   | <b>Holiday and Off-Day Management:</b> Define holidays and off-days. Classes scheduled on these dates are automatically cancelled or flagged.                                  | High     |
| AN-09   | <b>Student Semester Override:</b> Manually update a student's semester when required (e.g., semester drop or program withdrawal).                                              | Medium   |
| AN-10   | <b>Role Management (Student ↔ CR):</b> Promote a Normal Student to Class Representative or demote a CR back to a Normal Student.                                               | Medium   |
| AN-11   | <b>User Registration Verification:</b> Verify registration requests against pre-loaded institutional records before granting access.                                           | High     |
| AN-12   | <b>System-Wide Visibility:</b> Access all information related to users, schedules, semesters, and batches for monitoring and administration.                                   | Medium   |

### 2.3 Operating Environment

The system will operate in the following environment:

**Client-Side (Frontend):**

| Component            | Specification                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------|
| Platform             | Web browser (cross-platform)                                                                       |
| Supported Browsers   | Google Chrome (v100+), Mozilla Firefox (v100+), Microsoft Edge (v100+), Safari (v15+), Brave       |
| Supported Devices    | Desktop computers, laptops, tablets, and smartphones                                               |
| Responsive Design    | The UI will be fully responsive, adapting to screen sizes from 320px (mobile) to 1920px+ (desktop) |
| Internet Requirement | Active internet connection required for all operations                                             |

**Server-Side (Backend):**

| Component      | Specification                                                             |
|----------------|---------------------------------------------------------------------------|
| Runtime        | Node.js (LTS version, v18+) with TypeScript                               |
| Framework      | Express.js or equivalent Node.js HTTP framework                           |
| Database       | PostgreSQL (v14+)                                                         |
| ORM            | Prisma ORM for type-safe database access and migrations                   |
| Authentication | JWT (JSON Web Tokens) for session management; bcrypt for password hashing |
| Email Service  | SMTP-based email service for verification emails and notifications        |

**Deployment:**

| Component          | Specification                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Hosting (Frontend) | Vercel (Cloud hosting platform for static and serverless frontends)                                    |
| Hosting (Backend)  | Render (Cloud hosting platform for web services)                                                       |
| Database Hosting   | Neon DB (Serverless PostgreSQL database hosting)                                                       |
| Domain             | No custom domain; platform-provided default subdomains will be used (e.g., .vercel.app, .onrender.com) |

| Component | Specification                                                                                  |
|-----------|------------------------------------------------------------------------------------------------|
| SSL       | HTTPS enforced via platform-provided default SSL certificates (automated by Vercel and Render) |

### Tech Stack Summary:

| Layer          | Technology                        |
|----------------|-----------------------------------|
| Frontend       | React (v18+) with TypeScript      |
| Backend        | Node.js with TypeScript           |
| Database       | PostgreSQL + Prisma ORM           |
| Authentication | JWT + bcrypt + Email Verification |

## 2.4 Constraints

### 2.4.1 Design and Implementation Constraints

This section documents the conditions, standards, and policies that bound all design and implementation choices for Smart Department . Every technical decision must remain within the limits described below.

#### 2.4.1.1 Regulatory and Legal Constraints

Smart Department stores and processes personal data of students, teachers, and administrative staff of a public university. Compliance with the following legal frameworks is **mandatory**:

- **Bangladesh ICT Act, 2006** (amended 2009, 2013) – governs electronic records, digital transactions, and unauthorised access to computer systems. The system must not store, transmit, or process data in ways that violate this Act.
- **Bangladesh Digital Security Act, 2018** – regulates digital content, cybercrime, and data privacy in Bangladesh. Any feature involving public data disclosure or inter-system data transfer must be evaluated against this Act before implementation.
- **Jahangirnagar University IT Usage Policies** – the system must conform to JU’s internal policies on acceptable use, data handling, and IT infrastructure. Deployment on JU servers requires prior approval from the JU IT Division.

### 2.4.1.2 University and Departmental Policy Constraints

- The system must reflect the CSE Department’s academic structure: **Honors** (8 semesters, 4 years) and **Masters** (2 semesters, 1 year), with each semester lasting approximately **6 months** and two semesters per academic year.
- The **weekly off-days are Friday and Saturday**, in accordance with the Bangladesh national weekend convention and JU’s academic calendar. The working week runs **Sunday through Thursday**.
- The system must support the department’s practice of **promoting all students at semester-end by default**, including students who have failed courses. The Admin retains full authority to override this for individual cases with a written reason.
- **Two CRs per batch** must be supported – one male and one female – as required by departmental convention.
- Student academic records (marks, results, advisory signals) must be accessible only to the individual student, their assigned teachers, and the Admin. Cross-batch data visibility is not permitted for any non-admin user.

### 2.4.1.3 Language and Localisation Constraints

- The UI must fully support both **English** (default) and **Bangla**, switchable via a header toggle. All user-facing messages, validation errors, and notifications must be available in both languages.
- All date and time values must be stored and displayed in the **Asia/Dhaka** time-zone (UTC+6). Bangladesh does not observe Daylight Saving Time; no DST conversion logic is required.
- PDF exports (exam routines, result sheets) must render Bangla Unicode text correctly. Font selection for Bangla in PDF output is an open issue (Appendix D, OI-5) to be resolved during the UI/UX design phase.

### 2.4.1.4 Accessibility Constraints

- The system must meet **WCAG 2.1 Level AA** accessibility standards – a hard constraint driven by the presence of physically disabled students in the user base.

- Mandatory accessibility features include: full **keyboard navigation**, **ARIA labels** on all interactive elements, a minimum **colour contrast ratio** of 4.5:1 for normal text (3:1 for large text), and compatibility with common screen readers.
- Accessibility features must not be degraded or disabled by any user preference setting or UI theme.

#### 2.4.1.5 Security Constraints

- All client-server communication must be over **HTTPS (TLS 1.2 or higher)**. Plain HTTP must be permanently redirected to HTTPS; no mixed-content is permitted.
- Passwords must be hashed with **bcrypt** (minimum cost factor 12) or **argon2id**. Plain-text or reversibly encrypted passwords are strictly prohibited.
- The password policy enforces: minimum **8 characters**, at least one uppercase letter, one lowercase letter, one digit, and one special character.
- **Open self-registration is prohibited**. All user accounts must be created by the Admin (CSV bulk upload or manual entry) and require email verification before first login.
- **Role-Based Access Control (RBAC)** must be enforced on every server-side API endpoint. Client-side checks are supplementary only and are not a substitute for server-side authorisation.
- The system must protect against the **OWASP Top 10** vulnerabilities, including SQL injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF).
- Failed login attempts must be rate-limited: **maximum 5 attempts per 15 minutes** per account, followed by a 15-minute lockout.

#### 2.4.1.6 Technical and Platform Constraints

- **Database:** A relational DBMS is required – **PostgreSQL 14+** or **MySQL 8+**. NoSQL or non-relational primary data stores are not permitted in v1.
- **Server OS:** The production environment must run **Linux** (Ubuntu 22.04 LTS or later). Windows Server is not supported.

- **Web server:** **Nginx** or **Apache** is required as the reverse proxy. The application server technology (Node.js, Django, Laravel, etc.) is deferred to the design phase (Appendix D, OI-1).
- **Browser support:** The system must function correctly in the latest two stable releases of **Chrome**, **Firefox**, **Edge**, and **Safari**. Internet Explorer support is explicitly out of scope.
- **Responsive design:** The UI must be usable at a minimum width of **360 px**, with layouts optimised for **768 px** (tablet) and **1280 px** (desktop).
- **File storage:** Local disk storage is acceptable for v1. An S3-compatible object store is recommended for v2 when video file volumes may exceed local capacity.
- **No native mobile application** will be developed in v1. The web application must be mobile-responsive and functional on Chrome for Android and Safari for iOS.

#### 2.4.1.7 Scope Constraints – Out of Scope for v1

The following capabilities are explicitly **excluded from v1**:

- Financial transactions or fee payment processing.
- Integration with JU's central registration or examination systems (planned for v2 – Appendix D, OI-6).
- SMS notifications; email and in-app notifications only in v1 (SMS via carrier API planned for v2 – Appendix D, OI-4).
- Live video lectures or video conferencing features.
- A native Android or iOS application (Appendix D, OI-10).
- Automated Ramadan-aware scheduling; Admin may apply adjustments manually in v1 if required (Appendix D, OI-8).

## 2.5 Assumptions

- The university will provide correct and updated data of students, teachers, courses, and departments.

- Each user will have a unique ID for login and identification.
- Users are assumed to have basic knowledge of using web applications.
- A stable internet connection will be available for all users.
- Only authorized users (Admin, Teacher, CR) can perform system operations.
- The system will initially support a single department but can be extended in the future.
- The server and infrastructure will be sufficient to handle expected usage.

## 3. Requirements

### 3.1 Functional Requirements

The functional requirements are organized by module and written in User Story format with confirmation (acceptance criteria).

#### 3.1.1 Authentication & User Management

##### **FR-01: User Registration**

**As a** new user (Student, CR, Teacher, or Admin),

**I want to** register on the platform by providing my details,

**So that** I can create an account and access the system based on my role.

##### **Confirmation:**

- The registration form collects the user's full name, university ID, email address, password, role selection (Student/Teacher), batch (for students), and program type (Honours/Masters for students).
- Upon form submission, the system cross-verifies the provided university ID, email, and batch against the admin pre-loaded dataset.



- If verification fails, the system displays the error message: *"Your information does not match our records. Please contact the department admin."*
- If verification succeeds, the system sends a verification email containing a unique activation link to the provided email address.
- The account is created in an inactive state and is activated only after the user clicks the email verification link.
- Duplicate registration using the same university ID or email is rejected with an appropriate error message.

**FR-02: Email Verification**

**As a** newly registered user,

**I want to** verify my email address by clicking a verification link,

**So that** my account is activated and I can log in.

**Confirmation:**

- Upon successful registration, the system sends an email containing a unique verification link that expires after 24 hours.
- Clicking the verification link activates the user's account and redirects the user to the login page with a success message.
- If the verification link has expired, the system displays an error message and provides an option to resend the verification email.
- Accounts that remain unverified for 7 days are automatically removed from the system.

**FR-03: User Login**

**As a** registered and verified user,

**I want to** log in using my email and password,

**So that** I can access my personalized dashboard.

**Confirmation:**

- The login form accepts the user's email address and password.
- Upon successful authentication, the system issues a JWT token and redirects the user to their role-specific dashboard.

- If the credentials are invalid, the system displays the error message: *"Invalid email or password."*
- After five consecutive failed login attempts, the user's account is temporarily locked for 15 minutes.
- The issued JWT token expires after 24 hours, requiring the user to authenticate again.

**FR-04: Password Change**

**As a** logged-in user,

**I want to** change my password,

**So that** I can maintain the security of my account.

**Confirmation:**

- The user must provide their current password and the new password (with confirmation).
- The new password must meet minimum security requirements: at least 8 characters, including one uppercase letter, one lowercase letter, one digit, and one special character.
- If the current password is incorrect, the system displays an error.
- Upon successful password change, all existing sessions (JWT tokens) are invalidated, and the user is redirected to the login page.

**FR-05: Forgot Password**

**As a** user who has forgotten my password,

**I want to** reset my password via email,

**So that** I can regain access to my account.

**Confirmation:**

- The user enters their registered and verified email address on the forgot password page.
- The system sends a time-limited (5-minute expiry) OTP to the email address.
- The user enters the OTP, and upon successful validation, is redirected to a change password page.

- The user submits a new password (with confirmation).
- After resetting, all existing sessions are invalidated, and the user is redirected to the login page with a success message.
- If the email is not found in the system or is not verified, the system displays an error message immediately.

### 3.1.2 Batch & Semester Management

#### FR-06: Batch Creation

**As an Admin,**

**I want to** create a new batch by specifying the program, admission year, and section,  
**So that** students can be grouped correctly throughout their academic life.

#### Confirmation:

- The Admin provides the program (Honours/Masters), admission year, section, and expected student count.
- The system generates a unique batch ID in the format: `Program-Year-Section` (e.g., Honors-2024-A).
- Duplicate batches with the same program, year, and section are rejected.
- Newly created batches are initially assigned Semester 1 and marked as Active.

#### FR-07: Semester Creation & Activation

**As an Admin,**

**I want to** create and activate academic semesters,  
**So that** teaching activities can begin for the selected batches.

#### Confirmation:

- The Admin specifies the program, semester number, start date, and end date.
- The semester is initially created with status **Upcoming**.
- Upon activation, the system automatically generates academic calendars for all selected batches.
- The uploaded class routine becomes active.

- All students of the affected batches receive an in-app notification and email informing them that the semester has started.

**FR-08: Batch Promotion**

**As an Admin,**

**I want to** promote a batch to the next semester,

**So that** students progress through their academic program.

**Confirmation:**

- The system verifies that the batch has not already completed its final semester.
- The current semester is closed automatically.
- The next semester becomes the active semester.
- All students receive a notification regarding the semester promotion.
- Every promotion action is recorded in the Audit Log.

**FR-09: Student Batch Transfer**

**As an Admin,**

**I want to** transfer a student between batches or sections,

**So that** departmental administrative changes can be managed correctly.

**Confirmation:**

- The Admin selects the student, destination batch, and provides a mandatory transfer reason.
- The student is removed from the previous batch and assigned to the new batch.
- The student's academic history remains unchanged.
- The transfer action is recorded in the Audit Log with timestamp and administrator identity.

**3.1.3 Class Routine Management****FR-10: Class Routine Creation**

**As an Admin,**

**I want to** create or upload the class routine for each batch,

**So that** students and teachers can follow a structured weekly schedule.

**Confirmation:**

- The Admin can create routine entries manually or upload them using a CSV file.
- Each routine entry includes batch, semester, course, teacher, room, day, and time.
- The system validates all entries before saving.
- Successfully imported routines become available after semester activation.

#### **FR-11: Routine Conflict Detection**

**As the System,**

**I want to** detect scheduling conflicts before saving routine entries,

**So that** overlapping classes are prevented.

**Confirmation:**

- The system checks for room conflicts.
- The system checks for teacher conflicts.
- The system checks for batch conflicts.
- If any conflict exists, the routine entry is rejected with a descriptive error message.
- Only conflict-free entries are saved.

#### **FR-12: Routine Viewing**

**As a Student, Teacher, or Class Representative (CR),**

**I want to** view my class routine,

**So that** I know my scheduled classes.

**Confirmation:**

- Students can view only the routine of their own batch.
- CRs can view only the routine of their assigned batch.
- Teachers can view all classes assigned to them across different batches.
- The routine displays course name, teacher, room number, day, and time.
- Any approved routine update is immediately reflected in the dashboard.

### 3.1.4 Classroom & Resource Allocation

#### FR-13: Room Allocation

**As an Admin,**

**I want to** allocate classrooms, labs, and the multipurpose room to academic schedules,  
**So that** every class, lab session, and exam has an assigned room without conflicts.  
(Seminars and workshops are handled by the Chairman).

#### Confirmation:

- When creating or modifying any scheduled academic event (class, exam), the admin selects a room from the available pool: R-101, R-102, R-103, R-201, R-203, R-302, R-105, R-202.
- The system checks for conflicts: if the selected room is already occupied at the requested date and time, the allocation is blocked with an error: *"Room [R-XXX] is already allocated to [Event Name] from [Start Time] to [End Time] on [Date]."*
- The system displays a room availability matrix showing all rooms' schedules for the selected date, allowing the admin to visually identify open slots.
- Successfully allocated rooms appear in the routine and are visible to all affected users.

#### FR-14: Room Conflict Detection

**As the System,**

**I want to** automatically detect and prevent scheduling conflicts for rooms, teachers, and batches,

**So that** no two events overlap for the same resource at the same time.

#### Confirmation:

- Before any schedule creation or modification (routine generation, class rescheduling, CT scheduling, exam routine), the system checks for three types of conflicts:
  - **Room Conflict:** The target room is already occupied at the requested date and time.
  - **Teacher Conflict:** The teacher is already assigned to another class/event at the requested date and time.
  - **Batch Conflict:** The batch already has a class/event at the requested date and time.

- If any conflict is detected, the operation is blocked, and a detailed conflict message is displayed specifying the conflicting event, room, teacher, or batch. (Note: Seminar allocations have a special conflict resolution workflow handled by the Chairman).
- Conflict detection operates in real-time and applies to all users who can modify schedules (Admin, Teacher).

**FR-14B: Seminar/Workshop Allocation (Chairman)**

**As the** Chairman,

**I want to** allocate rooms and time slots for seminars and workshops,

**So that** departmental events can be scheduled dynamically.

**Confirmation:**

- The Chairman can select any room, date, and time slot to schedule a seminar or workshop.
- If the selected slot conflicts with an existing scheduled class (due to room unavailability or required student participation), the system will prompt the Chairman with the conflict details.
- The Chairman can then use the system to send an automated email notification to the respective course teacher(s) requesting them to cancel their class.
- Once the course teacher cancels their class, the room becomes available, and the Chairman can successfully allocate the seminar/workshop.
- Only the Chairman has the privilege to perform this action.

### 3.1.5 Class Update & Reschedule Management

**FR-15: Class Cancellation**

**As a** Teacher,

**I want to** cancel one of my scheduled classes,

**So that** students are informed and the room is freed up for other use.

**Confirmation:**

- The teacher can select any of their upcoming scheduled classes and cancel it.
- The system updates the class status to “Cancelled” in the routine.

- The cancelled class slot is freed, and the room becomes available for other allocations.
- Students of the affected batch see the cancellation reflected in their dashboard with a “Cancelled” label.
- The teacher can only cancel their own classes (verified via unique teacher ID).

**FR-16: Class Time Update**

**As a Teacher,**

**I want to** change the time of one of my scheduled classes on the same day,

**So that** I can adjust for personal or departmental needs.

**Confirmation:**

- The teacher selects a scheduled class and specifies a new time slot on the same day.
- The system performs conflict detection (room, teacher, batch) for the new time slot.
- If no conflict exists, the class time is updated, and the change is reflected on all users’ dashboards.
- If a conflict exists, the update is blocked with a descriptive error message.
- The original time slot is freed upon successful update.

**FR-17: Student-Initiated Cancellation Request**

**As a Student(CR),**

**I want to** request my teacher to cancel a scheduled class,

**So that** the teacher is aware of the need and can decide whether to cancel it.

**Confirmation:**

- The student(CR) can select any upcoming scheduled class of their batch and submit a cancellation request with a reason (e.g., “no faculty available,” “clash with another exam”).
- The request is sent to the specific teacher assigned to that class, along with the student’s name/batch and reason.
- The request status is set to “Pending” and is visible to the student on their dashboard.



- The teacher receives a notification and can view all pending requests from students for their classes.
- The teacher can Approve or Deny the request:
  - If Approved, the system triggers the existing FR-15 (Class Cancellation) flow — the class status updates to “Cancelled,” the room is freed, and all students of the batch see the “Cancelled” label.
  - If Denied, the class remains unchanged, and the student is notified with the teacher’s reason (optional) for denial.
- A student can only request cancellation for classes belonging to their own batch.
- Only one active (pending) cancellation request per student per class is allowed at a time — duplicate requests are blocked until the prior one is resolved.
- All request actions (submitted, approved, denied) are logged with timestamps for record-keeping.

**FR-18: Student-Initiated Reschedule Request**

**As a Student(CR),**

**I want to** request my teacher to reschedule a scheduled class,

**So that** the class can be moved to a more suitable time instead of being missed entirely.

**Confirmation:**

- The student(CR) can select an upcoming scheduled class and submit a reschedule request with a reason and, optionally, a preferred new date/time.
- The request is sent to the assigned teacher with status “Pending,” visible on the student’s dashboard.
- The teacher can review the request and:
  - Accept as proposed — triggering the existing FR-16 (same-day time update) or FR-19 (different-day reassignment) flow, depending on whether the proposed slot is same-day or a different day, including full conflict detection (room, teacher, batch).
  - Accept with modification — the teacher can adjust the proposed date/time/-room before confirming, subject to the same conflict detection.
  - Deny — the class remains unchanged, and the student is notified with the reason (optional).

- If accepted (as proposed or modified), the class appears on all affected users' dashboards with the "Rescheduled" indicator.
- A student can only request rescheduling for classes belonging to their own batch.
- Only one active reschedule request per student per class is allowed at a time.
- All request actions are logged with timestamps.

**FR-19: Class Reassignment to Another Day**

**As a Teacher,**

**I want to** reschedule one of my classes to a different day,

**So that** I can make up for cancelled classes or accommodate changes in my schedule.

**Confirmation:**

- The teacher selects a scheduled class and specifies a new date, time slot, and (optionally) a different room.
- The system performs comprehensive conflict detection for the new date, time, room, teacher, and batch.
- If no conflicts are found, the class is reassigned to the new slot. The system suggests available time slots and rooms for the selected date to assist the teacher.
- The rescheduled class appears in all affected users' dashboards with a "Rescheduled" indicator.
- The teacher can only reschedule their own classes.

### 3.1.6 Holiday & Off-Day Management

**FR-20: Holiday Declaration**

**As an Admin,**

**I want to** declare holidays and off-days in the academic calendar,

**So that** no classes are scheduled on those days and the system accurately reflects the department's working schedule.

**Confirmation:**

- The admin can add a holiday/off-day by specifying: date, reason/name (e.g., "Independence Day," "Eid-ul-Fitr," "University Foundation Day"), and scope (all batches or specific batches).

- When a holiday is declared, all classes scheduled on that date are automatically flagged as “Holiday — No Class.”
- If a holiday is declared after the routine has already been generated, the system retroactively marks all classes on that date as cancelled.
- Holidays are visible in all users’ calendar views with a distinct visual indicator.
- The admin can remove a previously declared holiday, which re-enables the classes for that date (if they were auto-cancelled, they are restored to their original status).

**FR-21: Teacher Request for Extra Class on a Holiday**

**As a Teacher,**

**I want to** request the Admin’s approval to hold an extra class on a declared holiday/off-day,

**So that** I can make up for lost instruction time when a genuine need arises.

**Confirmation:**

- The teacher can select a declared holiday date and submit a request specifying: batch, subject, proposed time slot, and room (optionally the system can suggest available rooms), and a reason.
- The system performs conflict detection (room, teacher, batch) for the proposed slot against any other approved extra classes already scheduled on that holiday, even though regular classes are suppressed on holidays.
- The request status is set to “Pending” and is visible to the teacher on their dashboard.
- The Admin receives a notification and can view all pending extra-class requests, along with holiday context (name/reason for the holiday and its declared scope).
- The Admin can Approve or Deny the request:
  - If Approved, the system adds the class to the routine for that date/time/-room, overriding the “Holiday — No Class” status only for that specific slot (the rest of the holiday remains unaffected for other batches/classes). The class appears on the teacher’s and students’ dashboards with an “Extra Class (Approved)” indicator.
  - If Denied, no changes are made to the routine, and the teacher is notified with the Admin’s reason (optional) for denial.

- The teacher can only request extra classes for their own assigned batches/subjects.
- If the Admin later removes the holiday itself (per FR-20), any approved extra-class requests tied to that date remain valid as regular scheduled classes, since the date is no longer a holiday.
- All request actions are logged with timestamps for audit purposes.

### 3.1.7 CT/Tutorial and Lab test Scheduling

#### FR-22A : CT Slot Scheduling

##### As a Teacher

**I want to** schedule a Class Test by replacing an existing class slot, **So that** *students are tested without creating an extra slot in the routine.*

##### Confirmation:

- Teacher selects an existing class slot from their assigned routine and specifies the room number.
- **Room rule :** CT/Tutorial is held in room **202** by default . If room 202 is unavailable due to a conflict, then suggest available 101, 102, 103 .
- Before confirming, the system performs comprehensive conflict detection for:
  - Room availability — checks CT, final exam, seminar, workshop, and approved extra classes at the same time in room 202.
  - Teacher availability.
  - Batch availability.
  - University declared holidays or off-days.
  - Outside official department working hours.
- **Two CTs (different courses) for the same batch fall on the same day:** the system does not reject, but warns the teacher and CR that another CT is already scheduled that day and requires the teacher to explicitly confirm before proceeding.
- If confirmed and all other checks pass: the original class slot is marked as replaced, the CT is saved, and students and CR are notified.

- Teacher can edit CT details before the scheduled date. Editing after the date is blocked.
- Teacher can cancel before the date; cancellation restores the original slot and notifies students and CR.

### **FR-22B    Lab Test slot Scheduling**

#### **As a Teacher,**

**I want to** schedule a Lab Test by replacing an existing lab class slot, **So that** *students are assessed in the lab without creating an extra slot.*

#### **Confirmation:**

- Teacher selects an existing lab class slot from their assigned routine
- **Room rule:** Lab Test must be held in room **203** or **302** only. Any other room input is rejected with: *“Lab Tests are only permitted in rooms 203 and 302.”*
- Before confirming, the system performs comprehensive conflict detection for:  
    **Room availability** , **Teacher availability** ,**Batch availability** ,University declared **holidays or off-days** .,Outside official department working hours (**9 AM to 5 PM**).
- **Check for lab equipment (PC , monitor , speaker) condition and usability**
- **Two Lab Tests (different courses) for the same batch on the same day:** system warns the teacher and CR and requires the teacher to explicitly confirm. Does not block.
- If confirmed and all checks pass: the original lab slot is marked as replaced, the Lab Test is saved, and students and CR are notified.
- Teacher can edit Lab Test details before the scheduled date. Editing after the date is blocked.
- Teacher can cancel before the date; cancellation restores the original slot and notifies students and CR.

### 3.1.8 Assignment Scheduling

#### FR-23A Assignment Creation & Publication

*As a Teacher,*

**I want to** *create and publish assignments with deadlines,*

**So that** *students are formally notified and have a clear submission window.*

**Confirmation:**

- Teacher provides: title, description, course, batch, deadline (date and time), and optional attachments.
- Teacher selects **submission mode**:
  - **Online** — students submit digitally. Accepted formats: PDF, PPT, PPTX, DOC, DOCX, ZIP, RAR, and other common formats.
  - **Offline** — students must submit physically.
- Teacher configures **Allow Late Submission** (on/off). If enabled, teacher sets a late submission cutoff date which must be after the original deadline.
- Deadline must be a future date. Rejected if in the past.
- On publication, students and CR are notified with title, deadline, submission mode, and late submission status.
- Teacher may edit or **extend the deadline on CR's request**
- After the deadline passes, all editing is blocked.

#### FR-23B Student Assignment Submission

*As a Student,*

**I want to** *submit my assignment,*

**So that** *my work is recorded.*

**Confirmation:**

- **Online submission:**

- Student uploads a file in an accepted format (PDF, PPT, PPTX, DOC, DOCX, ZIP, RAR, etc.).
  - Submission before deadline — recorded and labelled **On Time**.
  - Submission after deadline, late submission disabled — rejected. No record is created.
  - Submission after deadline, late submission enabled — accepted and labelled **Late Submitted** with exact timestamp.
- **Offline submission:**
    - Student marks submission as **Physically Delivered** on the system after handing it in.

### FR-23C Late Submission Request (CR)

*As a Class Representative (CR),*

**I want to** *request a deadline extension for my batch,*

**So that** *students who missed the deadline can still submit their work.*

#### **Confirmation:**

- CR submits a request with a reason and a proposed extended deadline after consulting with tw batch .
- Request is sent to the assigned teacher with status **Pending**. Visible on both the CR's and students' dashboards.
- Teacher can:
  - **Approve** — Expand submission deadline .
  - **Deny** — no changes are made. CR is notified with an optional reason.

## 3.1.9 Final Exam Routine Management

### FR-24 Exam Routine Creation

*As an Admin,*

**I want to** *create the final exam routine by defining exam slots per course and batch,*

**So that** *examinations are properly scheduled without conflicts.*

**Confirmation:**

- Admin provides for each slot: course, batch, date, start time, end time, room number, and 3-5 invigilator.
- **Room rule** : Final exams are held in room **202** only. Any other room input is rejected with: “*Final exams are only permitted in room 202.*”
- Before confirming, the system performs comprehensive conflict detection for:
  - Room availability — checks CT, Lab Test, other batch’s final exam, seminar, workshop, and approved extra classes in room 202 at the same time.
  - Teacher/invigilator availability, for final exam **4-6 invigilator** needed
  - Batch availability.
  - University declared holidays or off-days.
  - Approved seminars and workshops.
  - Outside official examination hours.
- Additional departmental exam rules enforced:
  - **Minimum gap between exams**: A configurable minimum gap (e.g., one/two full day) must be maintained between two final exams for the same batch. Rejected if the gap is less than the configured minimum.
  - **No exam outside official examination hours** — rejected if outside the defined window.
  - **No exam scheduled in the past** — rejected.
  - **No exam during declared university vacations or holidays**
  - **Room capacity check** — room 202 capacity must meet or exceed the enrolled student count. Rejected if insufficient.
- If all checks pass, the slot is saved as DRAFT.
- Admin can view a conflict report at any time listing all unresolved clashes before publishing.
- Admin can edit or delete any draft slot before publication.



**FR-25 Exam Routine Publication & Class Routine Lock**

*As an Admin,*

*I want to publish the finalised exam routine,*

*So that all users can view the official schedule*

**Confirmation:**

- Before publication, the system runs a final full-pass conflict check across all draft slots.
- On publication:
  - All exam slots are set to **PUBLISHED**.
  - The **class routine is locked** for that batches — any edit attempt is blocked.
  - **No new CTs or Lab Tests** can be scheduled for the batch after publication.
  - All enrolled students and CRs are notified.
  - Admin can cancel the any final exam prior to at least one day to exam if any unwanted situation arises.

**3.1.10 Resource Sharing Module****FR-26: Resource Upload by CR**

*As a Class Representative (CR),*

*I want to upload study resources (notes, slides, past papers, and reference materials) to the platform,*

*So that all students across all batches can access and download them.*

**Confirmation:**

- The CR can upload files (PDF, DOCX, PPTX, XLSX, PNG, JPG) along with metadata including resource title, course name, semester, year, and resource type (Notes, Slides, Past Paper, or Other).
- Uploaded resources are displayed on the public Resource Page and organized by year and semester.
- All users, including visitors if the page is public, can browse and download the uploaded resources.

- The CR can delete resources they have uploaded, while administrators can delete any resource.
- The maximum allowed file size is 50 MB per file.
- Only supported file formats (PDF, DOCX, PPTX, XLSX, PNG, and JPG) are accepted.

**FR-27: Resource Page (Public)**

**As a** student, teacher, administrator, or visitor,

**I want to** browse and download study resources organized by year and semester,

**So that** I can easily access academic materials for study and reference.

**Confirmation:**

- The Resource Page is publicly accessible without requiring user authentication.
- Resources are organized hierarchically as: Year → Semester → Course → Resource Type.
- Each resource entry displays the resource title, course name, semester, upload date, uploader (CR name), and file size.
- Users can search and filter resources by course name, semester, year, or resource type.
- Each resource includes a download button.
- The system tracks and displays the total number of downloads for each resource.

**3.1.11 Result Generation Module****FR-28: Result Generation System**

**Title:** Access Semester Results and Performance Summary

**User Story**

*As a student, I want to view my semester results and performance summary online, so that I can easily track my academic progress without waiting for manual result publication.*

**Description**

The system allows Class Representatives (CRs) or authorized teachers to upload and publish semester results. After publication, students can securely access their semester results through the dashboard, where they can view course-wise marks, GPA, and an overall academic performance summary.

**Confirmation:**

- Class Representatives (CRs) or authorized teachers can upload course-wise marks.
- The system automatically generates the final result sheet, including GPA and grade summary.
- Students can view:
  - Course-wise marks.
  - Total GPA and grade summary.
- Results are visible only after they have been officially published.
- Students have read-only access and cannot edit or modify their results.

### 3.1.12 Role-Wise Dashboard & Notification System

**FR-29: Schedule Dashboard & Notification System**

**Title:** View Personalized Class Schedule and Receive Notifications

**User Story**

*As a student, I want to view my personalized class schedule in a dashboard and receive automatic notifications for any changes, so that I can stay updated without missing classes or schedule updates.*

**Description**

The system provides a personalized dashboard where students can view their daily and weekly class schedules based on their batch, semester, and enrolled courses. Whenever a class is rescheduled, canceled, or a new class is added, the system automatically sends notifications through the in-app notification panel or email, ensuring that students are always informed about schedule updates.

**Confirmation:**

- Students can view their complete weekly class schedule in a dashboard.
- The schedule is automatically filtered based on the student's batch and semester.
- The system sends notifications when:
  - A class is rescheduled.
  - A class is canceled.
  - A new class is added.
- Notifications are displayed in the dashboard notification panel and may also be sent via email.

## 3.2 Non-Functional Requirements

### 3.2.1 Performance

- The system should load any page within **3 seconds** under normal usage.
- It should support at least **300 concurrent users** without significant performance degradation.
- Routine, result, and notice retrieval should be completed quickly.

### 3.2.2 Security

- Every user must log in using a unique username/email and password.
- Passwords must be stored securely using encryption (hashing).
- Users can only access features based on their assigned roles (Admin, Teacher, CR, Student).
- Important actions should be logged for auditing purposes.

### 3.2.3 Availability

- The system should be available **24/7**, except during scheduled maintenance.
- Regular database backups should be performed to prevent data loss.

### **3.2.4 Reliability**

- The system should maintain data consistency during unexpected failures.
- Data should be recoverable from backups when necessary.

### **3.2.5 Usability**

- The interface should be simple and user-friendly.
- The system should display clear error and confirmation messages.

### **3.2.6 Compatibility**

- The system should work on modern web browsers such as Chrome, Firefox, Edge, Safari, and Brave.
- It should support desktops, laptops, tablets, and mobile devices.

### **3.2.7 Scalability**

- The system should support future additions of batches, classrooms, laboratories, and new modules without major modifications.

### **3.2.8 Maintainability**

- The source code should be modular and well documented.
- Updates and bug fixes should be easy to implement.

### **3.2.9 Data Integrity**

- The system should prevent duplicate or invalid data.
- Routine and result information should remain accurate and consistent.

### 3.2.10 Backup and Recovery

- The database should be backed up regularly.
- Deleted or modified data should be recoverable when required.

### 3.2.11 Portability

- The system should be deployable on both local servers and cloud platforms.

### 3.2.12 Response Time

- Login requests should complete within **2 seconds**.
- Searching routines, notices, and results should take less than **2 seconds** under normal conditions.